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$$f(x) = \sin(\text{Bgcot } x)$$

$$\text{neem } y = \text{Bgcot } x$$

$$\text{dan is } \cot y = x \text{ en } y \in]0, \pi[$$

we kunnen hier de identiteit $\sin^2 y + \cos^2 y = 1$ gebruiken om de cotangens te vinden:

$$\text{we weten dat } \cot y = \frac{\cos y}{\sin y}$$

$$\text{dan is } \cot^2 y = \frac{\cos^2 y}{\sin^2 y}$$

$$\sin^2 y + \cos^2 y = 1$$

$$\frac{\sin^2 y}{\sin^2 y} + \frac{\cos^2 y}{\sin^2 y} = \frac{1}{\sin^2 y}$$

$$1 + \cot^2 y = \frac{1}{\sin^2 y}$$

$$\frac{1}{\sin^2 y} = 1 + \cot^2 y$$

$$\frac{1}{\sin^2 y} = 1 + x^2 \text{ (als } \cot y = x \text{ is } \cot^2 y = x^2)$$

$$\sin^2 y = \frac{1}{1 + x^2}$$

$$\sin y = \pm \frac{\sqrt{1}}{\sqrt{1 + x^2}}$$

$$= + \frac{1}{\sqrt{1 + x^2}} \text{ want } y \in]0, \pi[$$

$$\text{dus is } \sin(\text{Bgcot } x) = \frac{1}{\sqrt{1 + x^2}}$$

References